

JAVA PROGRAMMING LAB

Student Performance Analysis System

Experiment No.	02
Title	Student Performance Analysis System using Classes and Objects
Source File	StudentPerformance.java

Objective / Aim

To design and implement a Student Performance Analysis System in Java using classes and objects. The program demonstrates object-oriented programming concepts, arrays, constructors, methods, loops, conditional statements, and String methods to calculate total marks, average, highest mark, lowest mark, percentage, grade, result, and performance remark.

System Requirements

Hardware: Standard PC/Laptop; Intel Core i3/i5 processor or equivalent; minimum 4 GB RAM; minimum 500 MB free storage.

Software: Windows / Linux / macOS; JDK 17 or JDK 21; Eclipse / IntelliJ IDEA / NetBeans / VS Code; Command Prompt / Terminal.

Problem Statement

Develop a Java application to analyze the academic performance of a student. The system accepts roll number, name, and marks in five subjects, then calculates total, average, highest and lowest marks, percentage, grade, pass/fail result, and performance remark.

Algorithm / Logic Description

1. Create a Student class with roll number, name, and marks array.
2. Create a constructor to initialize student details.
3. Calculate total marks by adding all marks.
4. Calculate average using total marks divided by the number of subjects.
5. Find highest and lowest marks using Math.max() and Math.min().
6. Calculate percentage using total marks out of 500.
7. Determine grade using percentage ranges.
8. Format the name using trim() and toUpperCase().
9. Display the complete performance report.
10. In main(), accept input using Scanner, create a Student object, and call displayDetails().
11. Close the Scanner.

Class Structure

Student: rollNo, name, marks[]; methods calculateTotal(), calculateAverage(), findHighest(), findLowest(), calculatePercentage(), calculateGrade(), displayDetails().

StudentPerformance: contains main(), accepts input, creates the Student object, and displays the report.

Source Code

```
import java.util.Scanner;

class Student {
    int rollNo;
    String name;
    int[] marks;

    // Constructor
    Student(int rollNo, String name, int[] marks) {
        this.rollNo = rollNo;
        this.name = name;
        this.marks = marks;
    }

    // Calculate total marks
    int calculateTotal() {
        int total = 0;

        for (int mark : marks) {
            total += mark;
        }

        return total;
    }

    // Calculate average
    double calculateAverage() {
        return calculateTotal() / (double) marks.length;
    }

    // Find highest mark
    int findHighest() {
        int highest = marks[0];

        for (int mark : marks) {
            highest = Math.max(highest, mark);
        }

        return highest;
    }

    // Find lowest mark
    int findLowest() {
        int lowest = marks[0];

        for (int mark : marks) {
            lowest = Math.min(lowest, mark);
        }

        return lowest;
    }

    // Calculate percentage
    double calculatePercentage() {
        return (calculateTotal() / 500.0) * 100;
    }

    // Determine grade
    String calculateGrade() {
        double percentage = calculatePercentage();

        if (percentage >= 90)
            return "A+";
        else if (percentage >= 80)
            return "A";
        else if (percentage >= 70)
            return "B";
        else if (percentage >= 60)
            return "C";
        else if (percentage >= 50)
            return "D";
        else
            return "F";
    }

    // Display complete report
    void displayDetails() {
        String formattedName = name.trim().toUpperCase();

        double percentage = calculatePercentage();
        double roundedPercentage = Math.round(percentage * 100.0) / 100.0;
        String grade = calculateGrade();
    }
}
```

```

        System.out.println("\n===== STUDENT PERFORMANCE REPORT =====");
        System.out.println("Roll Number      : " + rollNo);
        System.out.println("Student Name   : " + formattedName);
        System.out.println("Name Length    : " + formattedName.length());

        System.out.println("Total Marks      : " + calculateTotal());
        System.out.println("Average Marks    : " + calculateAverage());
        System.out.println("Highest Marks    : " + findHighest());
        System.out.println("Lowest Marks     : " + findLowest());
        System.out.println("Percentage       : " + roundedPercentage + "%");
        System.out.println("Grade           : " + grade);

        if (percentage >= 50)
            System.out.println("Result          : PASS");
        else
            System.out.println("Result          : FAIL");

        if (percentage >= 90)
            System.out.println("Remark          : Outstanding Performance");
        else if (percentage >= 80)
            System.out.println("Remark          : Excellent Performance");
        else if (percentage >= 70)
            System.out.println("Remark          : Very Good Performance");
        else if (percentage >= 60)
            System.out.println("Remark          : Good Performance");
        else if (percentage >= 50)
            System.out.println("Remark          : Satisfactory Performance");
        else
            System.out.println("Remark          : Needs Improvement");
    }
}

public class StudentPerformance {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Roll Number: ");
        int rollNo = sc.nextInt();

        sc.nextLine();

        System.out.print("Enter Student Name: ");
        String name = sc.nextLine();

        int[] marks = new int[5];

        System.out.println("Enter marks for 5 subjects:");

        for (int i = 0; i < 5; i++) {
            System.out.print("Subject " + (i + 1) + ": ");
            marks[i] = sc.nextInt();
        }

        Student student = new Student(rollNo, name, marks);

        student.displayDetails();

        sc.close();
    }
}

```

Compilation & Execution

Save the source code exactly as:

```
StudentPerformance.java
```

Compile:

```
javac StudentPerformance.java
```

Run:

```
java StudentPerformance
```

Sample Input

```
Enter Roll Number: 101
Enter Student Name: Neha
Enter marks for 5 subjects:
Subject 1: 95
Subject 2: 88
Subject 3: 92
Subject 4: 85
Subject 5: 90
```

Sample Output

```
===== STUDENT PERFORMANCE REPORT =====
Roll Number      : 101
Student Name     : NEHA
Name Length      : 4
Total Marks      : 450
Average Marks    : 90.0
Highest Marks    : 95
Lowest Marks     : 85
Percentage       : 90.0%
Grade            : A+
Result           : PASS
Remark           : Outstanding Performance
```

Important Concepts Used

Class and Object • Constructor • Array • Methods • Enhanced for loop • Math.max() • Math.min() • String trim(), toUpperCase(), length() • if-else • Scanner input

Result

The Student Performance Analysis System was successfully implemented in Java using classes, objects, constructors, arrays, methods, loops, conditional statements, and String operations. The program successfully calculates and displays the student's total, average, highest mark, lowest mark, percentage, grade, result, and performance remark.